



RAPHAEL PELERIN

AI internship, end of studies · from March 2027

I am looking for an end-of-studies internship in AI — the kind where I **add something to the technology rather than only consume it**. Calling an API is the easy part; the interface on top I can already build. I want the engineering in between. Five projects follow, four of them used by people who are not me.

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PORTFOLIO

portfolio-raphael-pelerin.vercel.app

LANGUAGES

FR · EN bilingual · 中文 HSK3

I am looking for an end-of-studies internship in AI. The kind where I add something to the technology rather than only consume it.

A
 —
 AVAILABILITY

●
OPEN TO OFFERS

AI internship, end of studies · 4–6 months
Starting March 2027

B
 —
 LANGUAGES

French
English
Mandarin
Italian

NATIVE
BILINGUAL
HSK 3
BASIC

C
 —
 EDUCATION

2022 – 2027	Epitech Technology Master's, Computer Engineering · Nice Functional and object-oriented programming, Unix, assembly, systems and network programming. Specialisations in DevOps, AI, web and mobile, and cybersecurity.
2025 – 2026	Beijing Jiaotong University Fourth year abroad · Beijing, China Software engineering, alongside Mandarin to HSK 3.
2022	Lycée Simone Veil General Baccalaureate Computer Science (NSI) and Mathematics.

D
 —
 EARLIER

2023	ARD Mobile development intern An Angular application to manage NFC cards and user profiles for alarm systems, including card scanning to display cardholder information.
2022	Intel · AI for Youth Competition – 2nd place, team Computer vision and object recognition in Python with TensorFlow.

E
 —
 CAPABILITIES

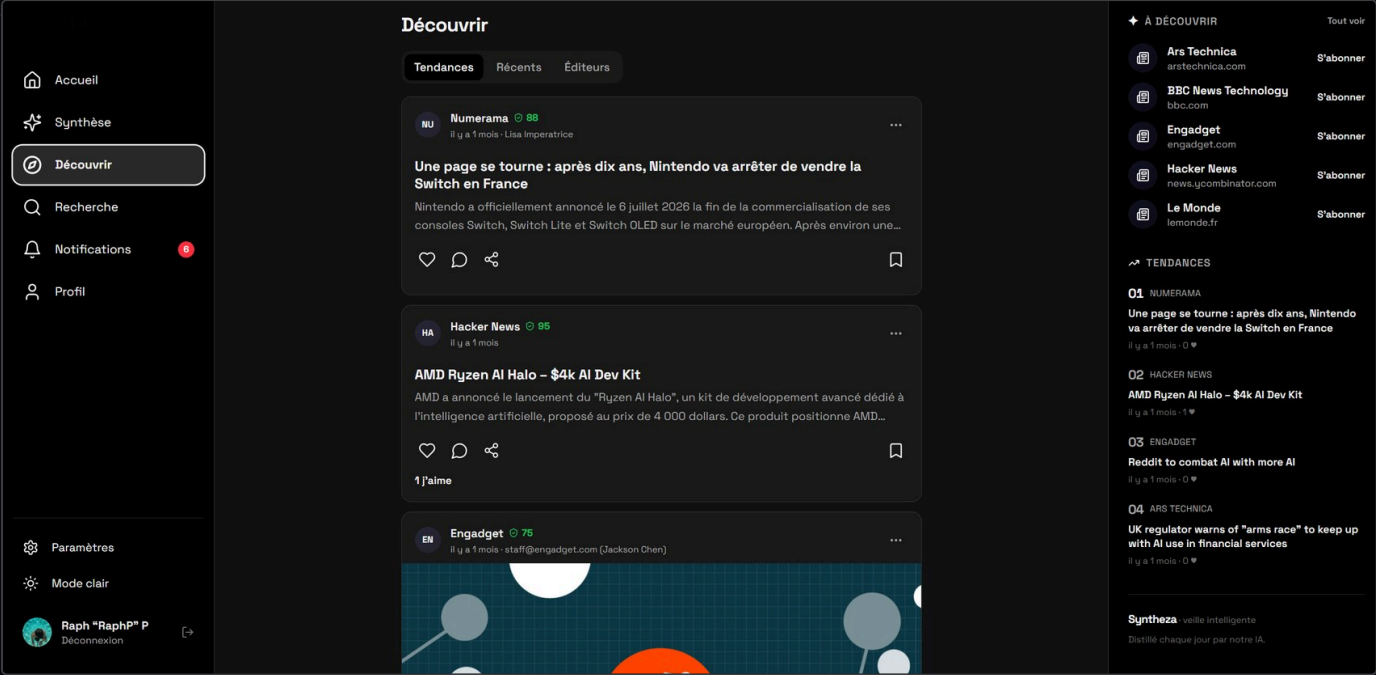
01	Web & mobile	React · React Native · Angular · Ionic · Node.js · Express · Android · iOS
02	Backend	FastAPI · Flask · Express · REST APIs · SOAP / XML
03	AI & generative	OpenAI · Anthropic Claude · Gemini · LLM integration · TensorFlow · diffusion models
04	Data & infra	PostgreSQL · MongoDB · MySQL · Prisma · Docker · containerisation · GitLab CI
05	Systems & audio	C++ · JUCE · real-time DSP · Unix · assembly · systems and network programming

01 — LEAD FRONTEND WEB · EPITECH G-EIP-700 · 2025 — 2026

SYNTHEZA

Strategic watch, against information overload

LIVE syntheza.ovh SOURCE github.com/Syntheza-DEV

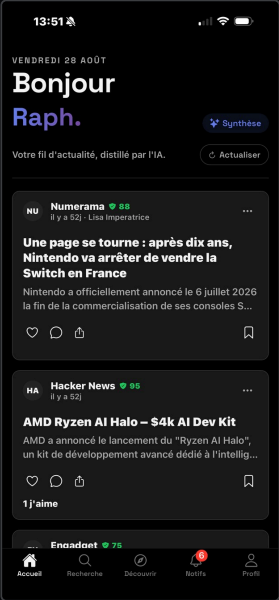


WEB CLIENT · DISCOVER FEED, PUBLISHERS, TRENDING AND TRUST FACTOR SCORES

A one-year entrepreneurial project run by three Epitech students working from three different countries. Syntheza ingests RSS at scale, deduplicates what it finds, and returns a daily synthesis instead of another feed. I lead the **web client** — the interface, the design system and the React component layer.

The green shield on every source is the **Trust Factor**: a corroboration score, so a claim made once does not read like one confirmed by five.

The team ran a field study of **20 professionals before writing code** — 85% reported daily overload, 70% lose one to two hours a day sorting noise. That study, not a hunch, set the scope.



MOBILE · EXPO

3

DEVELOPERS,
THREE COUNTRIES

~330 kB

GZIPPED
WEB BUNDLE

7

LANGUAGES
ON MOBILE

Live

IN PRODUCTION

- React 19
- TypeScript
- Tailwind v4
- Radix UI
- Framer Motion
- Vite
- Node 20
- Express
- Prisma 7
- PostgreSQL 15
- Gemini 2.5 Flash
- React Native · Expo

02 — SOLO PROJECT

FORGE

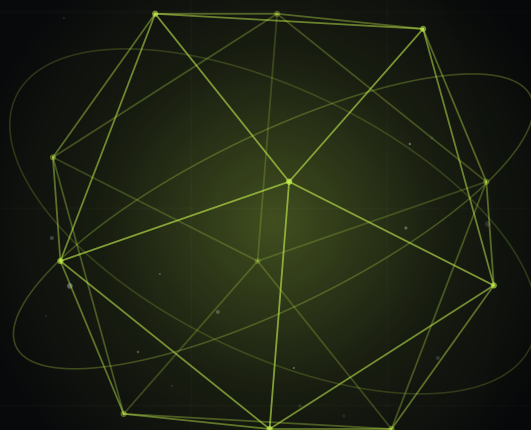
Game-asset generation that never leaves the machine

SOURCE github.com/RaphaelPelerin/forge-asset-generator

Generating *one* good image with an AI tool is easy. Generating **forty that look like they came from the same game** is the actual problem — and it is a software engineering problem, not a prompting one.

Forge answers it **as a service**: a FastAPI backend driving SDXL on a local GPU, so a game pipeline can call it like any other endpoint. Behind it are **recipes** — presets that own the style, resolution, sampler and post-processing for a category of asset. The user supplies only the subject, so two characters generated six months apart go through an identical pipeline and match.

It runs entirely on your own hardware. No subscription, no account, and **no image ever leaves the machine**. Nine-slice UI panels are not trusted to come out symmetrical — the tool keeps one corner, mirrors it, and *measures* the border thickness by scanning inward until the colour goes flat.



FOUR THINGS IT TOOK GENERATING IMAGES TO FIND

- 01 The text encoder only reads English. French subjects failed **6 out of 6**; English succeeded 6 out of 6. Fixed with a local translation model.
- 03 Style strength has to vary **by asset type**: pushed hard, icons collapse into unreadable clusters at their real 48 px display size.

- 02 The **75-token prompt budget overflows silently** — the end is dropped with no error. A test now stops any recipe exceeding it.
- 04 On 8 GB of VRAM an extra refiner pass took generation from **21 seconds to over 9 minutes**. Disabled by default, with the measured figure.

~4,100LINES OF
PYTHON + JS**59**TESTS, ~5 S,
NO GPU NEEDED**~22 s**PER GENERATED
IMAGE**0**EXTERNAL API CALLS
AT RUNTIME

PyTorch 2.13

CUDA 12.6

diffusers · SDXL

ControlNet

LoRA

FastAPI

Pydantic v2

pytest

React 18

Ant Design 5

Vite 6

03 — SOLO PROJECT · 2025

KEYS

A VST3 sampler, drawn by hand

SOURCE github.com/RaphaelPelerin/keys-gpt-vst

KEYS · SAMPLE LOADED, LIVE ADSR CURVE AND FILTER RESPONSE

A single-sample VST3 instrument built on JUCE. It pitch-shifts one sample across the keyboard through an ADSR envelope, a three-mode filter whose cutoff can follow that envelope, and forward or ping-pong looping across **1 to 16 voices**.

The interface uses **no GUI library at all** — every knob, curve and meter above is drawn with the JUCE Graphics API. The envelope view shows the ADSR shape with a live line tracking the playing voice; the filter graph draws the real frequency response with separate markers for the static and the envelope-modulated cutoff. The waveform is held as a **min/max envelope rather than raw samples**, so it stays sharp at any zoom, and the trim and loop markers are dragged directly on it.

16SIMULTANEOUS
VOICES**3**FILTER MODES,
ENVELOPE-MODULATED**0**EXTERNAL GUI
LIBRARIES**VST3**WINDOWS X64,
FULLY AUTOMATABLE

C++

JUICE 8

Real-time DSP

IIR filters

Projuicer

Visual Studio 2022

KEYS GPT

Describe a part, get a MIDI file

SOURCE github.com/RaphaelPelerin/keys-gpt

Thought for 4.6s

Here is a 4-bar boombap groove at 82 BPM — 54 hits across 4 lanes, written to the General MIDI percussion channel. Play it below, or download the MIDI and open it in your DAW.

PART	GROOVE	TEMPO	BARS	HITS	LANES	LENGTH
Drums	boombap	82 BPM	4	54	4	0:12

0:00 / 0:12

boom-bap-drum-groove-4-bars-s...

Raw response 718 tokens

GENERATED 4-BAR BOOMBAP GROOVE · 54 HITS, FOUR LANES, PLAYABLE IN THE BROWSER

You describe a part — a mood, a key, a tempo, a length — and get a piano roll you can hear and a MIDI file you can keep. It writes melodies, chord progressions, counter-melodies and drum grooves, and the part type is inferred from the prompt rather than picked from a menu.

It runs against the **OpenAI API** — gpt-4o-mini above — and the part is playable in the browser before anything is downloaded, so you hear it before deciding whether to keep it.

Lo-fi study melody in D minor, 84 bpm, 8 bars

Thought for 4.5s

Here is an 8-bar lo-fi melody in D dorian at 84 BPM — 35 notes over a seventh-chord bed. Play it below, or download the MIDI and open it in your DAW.

PART	KEY	TEMPO	BARS	NOTES	RANGE	LENGTH
Melody	D dorian	84 BPM	8	35	D4-F5	0:23

0:00 / 0:23

lo-fi-study-melody-in-d-minor...

Raw response 585 tokens

8-BAR MELODY · D DORIAN, 35 NOTES

5
PART TYPES,
INFERRED FROM PROMPT

MIDI
DOWNLOADABLE,
OPENS IN ANY DAW

OpenAI
API
COMPATIBLE

- React
- TypeScript
- OpenAI API
- Web Audio
- MIDI

05 — FRONT-END DEVELOPMENT INTERN · ANTIBES · APRIL — AUGUST 2025

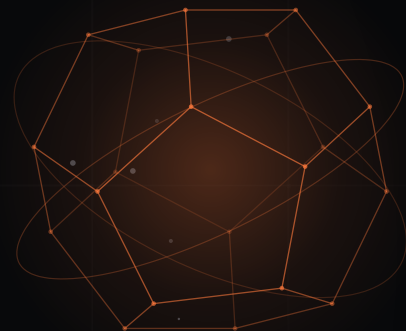
APPOLONIA

Casino floor management, where a mistake takes a machine offline

LIVE www.appolonia.fr/index.php/fr

Appolonia builds the systems casinos use to run their gaming floors. I worked on **KAMA**, the manager operators use to configure and deploy to **Kalypse Touch** terminals — the screens embedded in slot machines.

My responsibilities went from a dashboard widget, to redesigning whole modules, to building features from nothing.



REDESIGN

KT Updater

The module worked but was hard to use, and deployments failed often. I rebuilt it as three explicit steps — upload the file, choose whether it triggers a full reboot or an interface restart, then pick the devices and the schedule. Drag-and-drop uploads, live progress, and options that only appear when they apply.

In real deployments; used regularly by casino staff.

BUILT FROM SCRATCH

Administrator panel

Power policies for groups of terminals — "shut down every Friday at midnight", "lock from 2 AM to 6 AM daily" — with manual override for when the floor disagrees with the schedule.

BUILT FROM SCRATCH

Email update tool

Client update mails were written from scratch every time. A structured form now generates them in one consistent format, with a preview before sending.

ALSO

The "Today's Campaigns" dashboard widget, plus ongoing bug fixing, responsive layout work and component refactoring — with notes left in the code explaining why.

Angular 13

Ionic 6

TypeScript

SOAP / XML APIs

GitLab

F — AWAY FROM THE SCREEN

Skiing and taekwondo. Motorsport. Music production in a DAW — which is where the sampler came from. Travel, most recently a year of it in Beijing.

Full driving licence.

G — AVAILABILITY

● OPEN TO OFFERS

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Starting March 2027

GET IN TOUCH

LET'S BUILD SOMETHING SOLID

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BASED IN
Biot, France · 06